

Rounding to a given accuracy

A Cambridge insert: which digit decides, and what a rounded answer really claims.

LESSON 58

1 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 3.2

LESSON CLOCK

10'

12'

12'

6'

The deciding digit	10 min
Decimal places	12 min
Significant figures	12 min
Estimating	6 min

LEARNING OBJECTIVES

- Round to the nearest ten, hundred and thousand.
- Round to a given number of decimal places.
- Round to a given number of significant figures.
- Estimate a calculation by rounding first.

TEXTBOOK

National	pp. 164–166
Cambridge	Stage 7, pp. 36–40
Workbook	Exercise 3.2

01

Explanation

The deciding digit

Look at the digit **immediately after** the place you are rounding to. If it is 5 or more, round up; otherwise leave the digit alone.

NUMBER	TO THE NEAREST TEN	HUNDRED	THOUSAND
3 472	3 470	3 500	3 000
2 851	2 850	2 900	3 000
795	800	800	1 000

ONLY ONE DIGIT DECIDES

For 3 472 to the nearest hundred, the deciding digit is the 7, not the 2. Looking further along and “rounding twice” gives the wrong answer.

Decimal places

NUMBER	1 D.P.	2 D.P.	WHOLE NUMBER
3.472	3.5	3.47	3
0.0851	0.1	0.09	0
9.96	10.0	9.96	10
2.345	2.3	2.35	2

9.96 TO ONE DECIMAL PLACE IS 10.0, NOT 10

The rounding can carry into the next column, and the zero must stay to show the accuracy claimed. Writing 10 claims a whole number; 10.0 claims a tenth.

Significant figures

Significant figures are counted from the first non-zero digit, whatever the size of the number.

NUMBER	1 S.F.	2 S.F.	3 S.F.
3 472	3 000	3 500	3 470
0.0851	0.09	0.085	0.0851
205.6	200	210	206

LEADING ZEROS ARE NOT SIGNIFICANT; ZEROS BETWEEN DIGITS ARE

In 0.0851 the first significant figure is the 8. In 205.6 the 0 counts, because it sits between two significant digits.

Estimating

CALCULATION	ROUNDED	ESTIMATE	EXACT
$38 \cdot 21$	$40 \cdot 20$	800	798
$4\,187 \div 21$	$4\,000 \div 20$	200	199.4
$6.9 \cdot 5.2$	$7 \cdot 5$	35	35.88
$297 + 405$	$300 + 400$	700	702

AN ESTIMATE CATCHES THE BIG MISTAKES

It will not tell you whether the answer is 798 or 799, but it tells you at once that 7 980 is wrong. Estimating first is how a calculator error is caught.

02 Worked examples

EXAMPLE 1 Round 3 472 to the nearest hundred and to 2 significant figures.

- 1

Nearest hundred: the deciding digit is 7, so round up.
3 500
- 2

Two significant figures: the first two digits are 3 and 4; the next is 7.
3 500

ANSWER
3 500 both times

EXAMPLE 2 Round 9.96 to one decimal place.

- 1 The deciding digit is the 6.
- 2 Round the 9 up: it becomes 10 and carries.
- 3 10.0
The zero shows the accuracy claimed.

ANSWER

10.0

EXAMPLE 3 Estimate $4\,187 \div 21$.

- 1 Round to one significant figure each: $4\,000 \div 20$.
- 2 = 200
- 3 The exact value is 199.4 — close enough to check by.

ANSWER

About 200

03 Interactive model

Ask for an estimate before every calculation for a week; the habit catches more errors than any amount of rechecking.

Which digit decides?

3 472 to the nearest hundred:

3 400

3 500

3 470

3 000

2 851 to the nearest thousand:

2 000

3 000

2 900

2 800

3.472 to 1 d.p.:

3.4

3.5

3.47

4

9.96 to 1 d.p.:

9.9

10.0

10

9.96

0.0851 to 2 s.f.:

0.08

0.085

0.09

0.0851

205.6 to 2 s.f.:

200

210

205

206

Estimate $38 \cdot 21$:

600

800

1000

80

Estimate $6.9 \cdot 5.2$:

30

35

42

12

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To round	Yaxlitlash	Округлить
Nearest	Eng yaqin	Ближайший
Decimal place	O'nli xona	Десятичный знак
Significant figure	Muhim raqam	Значащая цифра
Deciding digit	Hal qiluvchi raqam	Решающая цифра
Estimate	Taxminiy hisob	Оценка
Accuracy	Aniqlik	Точность
Approximately	Taxminan	Приблизительно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A digit of 5 or more means:

round down

round up

leave it

add one to the last digit shown

795 to the nearest ten:

790

800

700

750

3.472 to 2 d.p.:

3.47

3.48

3.5

3.4

In 0.0851 the first significant figure is:

0

8

5

1

$3\,472$ to 1 s.f.:

3 000

3 400

4 000

3 500

An estimate is used to:

get an exact answer

catch a large error

avoid working

round twice

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 3 472 to the nearest ten

ANSWER 3 470

2. 3 472 to the nearest hundred

ANSWER 3 500

3. 2 851 to the nearest thousand

ANSWER 3 000

4. 795 to the nearest ten

ANSWER 800

5. 3.472 to 1 d.p.

ANSWER 3.5

6. 3.472 to 2 d.p.

ANSWER 3.47

7. 2.345 to 2 d.p.

ANSWER 2.35

Medium · the standard question

7 PROBLEMS

1. 9.96 to 1 d.p.

ANSWER 10.0

2. 0.0851 to 2 s.f.

ANSWER 0.085

3. 0.0851 to 1 s.f.

ANSWER 0.09

4. 205.6 to 2 s.f.

ANSWER 210

5. 3 472 to 1 s.f.

ANSWER 3 000

6. Estimate $38 \cdot 21$

ANSWER 800

7. Estimate $297 + 405$

ANSWER 700

Hard · stretch and reason

7 PROBLEMS

1. Estimate $4\,187 \div 21$

ANSWER 200

2. Estimate $6.9 \cdot 5.2$

ANSWER 35

3. 205.6 to 3 s.f.

ANSWER 206

4. Why is 10.0 not the same claim as 10?

ANSWER It claims accuracy to a tenth

5. The smallest whole number that rounds to 700 to the nearest hundred

ANSWER 650

6. The largest whole number that rounds to 700 to the nearest hundred

ANSWER 749

7. A length given as 4.5 m to 1 d.p. lies between

ANSWER 4.45 and 4.55 m

07 Homework

Homework — 5 tasks

Underline the deciding digit before you round anything.

1. Round 6 483 to the nearest ten, hundred and thousand.
2. Round 5.278 to 1 and to 2 decimal places.
3. Round 0.04672 to 1 and to 2 significant figures.
4. Estimate $58 \cdot 42$ and $8\,130 \div 19$.
5. Between which two values does a mass given as 3.2 kg to 1 d.p. lie?